Topic D

Multiple choice questions

1 The gravitational force between two point masses is F. What is the force when both masses are doubled and their separation halved?

A F

B $4F$

C $8F$

D $16F$

2 Two identical solid steel spheres touch. The gravitational force between them is F. The spheres are replaced by two touching solid steel spheres of double the radius. What is the force between the spheres now?

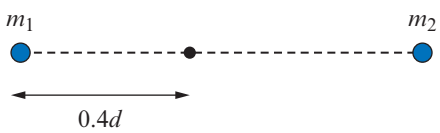
A $\frac{F}{4}$

B $\frac{F}{16}$

C $4F$

D $16F$

3 Two particles of masses m_1 and m_2 are a distance d apart.



The gravitational field strength on the line joining the particles at a distance of $0.4d$ from m_1 is zero. What is the ratio $\frac{m_1}{m_2}$?

A $\frac{2}{3}$

B $\frac{4}{9}$

C $\frac{2}{5}$

D $\frac{3}{5}$

4 The gravitational field strength on the surface of Earth is g. A planet has double the mass and double the radius of Earth. What is the gravitational field strength on the surface of the planet?

A $\frac{g}{4}$

B $\frac{g}{2}$

C g

D $2g$

5 A planet has four times the mass of the Earth, but the acceleration of gravity on the surface is the same as that on Earth. What is the radius of the planet in terms of the Earth radius R?

A $\frac{R}{4}$

B $\frac{R}{2}$

C $2R$

D $4R$

6 A satellite is in a circular orbit around a planet. The orbit radius is 5000 km, and the speed of the satellite is 2.0 km s^{-1}. What is the gravitational field strength at the position of the satellite?

A 0.8 N kg^{-1}

B 4.0 N kg^{-1}

C 8.0 N kg^{-1}

D Cannot be determined without knowing the mass of the planet.

7 The mass of a landing module on the Moon is 2000 kg. The gravitational field strength on the Moon is one-sixth that on Earth. What is the weight of the landing module on Earth?

A 330 N

B 2000 N

C 12 000 N

D 20 000 N

8 A planet has double the radius of Earth and half its density. What is the gravitational field strength on the surface of this planet?

A 5.0 N kg^{-1}

B 10 N kg^{-1}

C 20 N kg^{-1}

D 40 N kg^{-1}

9 A planet has radius R. The gravitational field strength at a height R from the surface is g.
What is the gravitational field strength at a point whose height from the planet's surface is $2R$?

A $\frac{g}{4}$

B $\frac{4g}{9}$

C $\frac{g}{2}$

D $\frac{2g}{3}$

10 A satellite orbits the Earth in a circular orbit. The only force on the satellite is the gravitational force from the Earth. Which of the following is correct about the acceleration of the satellite?

A It is zero.

B It is constant in magnitude and direction.

C It is constant in magnitude but not in direction.

D It is not constant in magnitude or direction.

11 A satellite is in a circular orbit around the Earth. The satellite enters an orbit further away from the surface. What change took place?

A The satellite's speed decreased.

B The satellite's speed increased.

C The satellite's mass decreased.

D The satellite's mass increased.